

Anxiety and Depression Symptoms After the *Dobbs* Abortion Decision

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IMPORTANCE In 2022, the US Supreme Court abolished the federal right to abortion in the *Dobbs v Jackson Women's Health Organization* decision. In 13 states, abortions were immediately banned via previously passed legislation, known as trigger laws.

OBJECTIVE To estimate changes in anxiety and depression symptoms following the *Dobbs* decision among people residing in states with trigger laws compared with those without them.

DESIGN, SETTING, AND PARTICIPANTS Using the nationally representative repeated cross-sectional Household Pulse Survey (December 2021-January 2023), difference-in-differences models were estimated to examine the change in symptoms of depression and anxiety after *Dobbs* (either the June 24, 2022, *Dobbs* decision, or its May 2, 2022, leaked draft benchmarked to the baseline period, prior to May 2, 2022) by comparing the 13 trigger states with the 37 nontrigger states. Models were estimated for the full population (N = 718 753), and separately for 153 108 females and 102 581 males aged 18 through 45 years.

EXPOSURE Residing in states with trigger laws following the *Dobbs* decision or its leaked draft.

MAIN OUTCOMES AND MEASURES Anxiety and depression symptoms were measured via the Patient Health Questionnaire-4 (PHQ-4); range, 0-12; scores of more than 5 indicate elevated depression or anxiety symptoms; minimal important difference unknown).

RESULTS The survey response rate was 6.04% overall, and 87% of respondents completed the PHQ-4. The population-weighted mean age was 48 years (SD, 17 years), and 51% were female. In trigger states, the mean PHQ-4 scores in the baseline period and after the *Dobbs* decision were 3.51 (95% CI, 3.44 to 3.59) and 3.81 (95% CI, 3.75 to 3.87), respectively, and in nontrigger states were 3.31 (95% CI, 3.27 to 3.34) and 3.49 (95% CI, 3.45 to 3.53), respectively. There was a significantly greater increase in the mean PHQ-4 score by 0.11 (95% CI, 0.06 to 0.16; $P < .001$) in trigger states vs nontrigger states. From baseline to after the draft was leaked, the change in PHQ-4 was not significantly different for those in trigger states vs nontrigger states (difference-in-differences estimate, 0.09; 95% CI, -0.03 to 0.21; $P = .15$). From baseline to after the *Dobbs* opinion, there was a significantly greater increase in mean PHQ-4 scores for those in trigger states vs nontrigger states among females aged 18 through 45 years (difference-in-differences estimate, 0.23; 95% CI, 0.08 to 0.37; $P = .002$). Among males aged 18 through 45 years, the difference-in-differences estimate was not statistically significant (0.14; 95% CI, -0.08 to 0.36; $P = .23$). Differences in estimates for males and females aged 18 through 45 were statistically significant ($P = .02$).

CONCLUSIONS AND RELEVANCE In this study of US survey data from December 2021 to January 2023, residence in states with abortion trigger laws compared with residence in states without such laws was associated with a small but significantly greater increase in anxiety and depression symptoms after the *Dobbs* decision.

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On June 24, 2022, the US Supreme Court overturned *Roe v Wade* and *Planned Parenthood v Casey* in the *Dobbs v Jackson Women's Health Organization* decision, abolishing the federal right to abortion. The decision came after a draft version was leaked to the public on May 2, 2022. *Dobbs* returned the issue of abortion regulation to state legislatures after nearly 50 years of federal protection, granting states nearly unrestricted power to regulate and ban abortions.

Prior to the opinion, the landscape of abortion laws in the US varied across states within the limits established by *Roe* and *Casey*. Thirteen states had passed trigger laws, anticipatory bans that would go into effect should *Roe* be overturned. These states include Arkansas, Idaho, Kentucky, Louisiana, Missouri, Mississippi, North Dakota, Oklahoma, South Dakota, Tennessee, Texas, Utah, and Wyoming.¹

Nascent research on the potential impacts of *Dobbs* raises concern that the overturning of *Roe* could harm population mental health.^{2,3} Research conducted prior to *Dobbs* established that individual abortion denial was significantly associated with adverse outcomes, including symptoms of depression and anxiety.^{4,5} Whether policies inhibiting abortion access have population-level effects on mental health is unclear. This study was designed to estimate the incremental change in depression and anxiety symptoms in trigger states vs nontrigger states, before and after the *Dobbs* decision. We hypothesized that residents of states with trigger laws would experience greater increases in symptoms of anxiety and depression than residents of states without trigger laws following the *Dobbs* decision. Additionally, we hypothesized that females younger than 45 years, an age cutoff used to compute lifetime abortion incidence, would experience the greatest increases in these symptoms.⁶

Methods

Ethics Statement

The Johns Hopkins Bloomberg School of Public Health Institutional Review Board deemed this study exempt. Informed consent was not required because the data were deidentified.

Study Design and Data Source

Thirteen waves of data from the US Census Bureau's Household Pulse Survey (HPS) public-use files, spanning December 29, 2021, to January 19, 2023, were used to construct the analytic sample. These waves were selected to avoid spillovers from the COVID-19-related policies that were in effect during 2020 and 2021 and to use the most recent data available at the time of the analysis. The HPS is a nationally representative, repeated cross-sectional survey of US households designed to provide near real-time information on the nation's well-being during the COVID-19 pandemic.^{7,8} The survey has been fielded by the Census Bureau approximately every 2 weeks since April 2020. The Census Contact Frame and Master Address File, databases that link individuals living in a household to their contact information, were used to reach respondents via text and email and to verify their home address. Upon contact, the HPS was then administered via

Key Points

Question Were there differences in the changes in depression and anxiety symptoms after the US Supreme Court overturned *Roe v Wade* among residents of states with trigger abortion bans compared with residents of states without such bans?

Findings In this retrospective analysis of survey data from 718 753 participants, residents of states that had passed trigger abortion bans experienced a significantly greater worsening of anxiety and depression symptoms than residents of states without such bans after the *Dobbs* decision compared with a baseline period before its draft was leaked (mean Patient Health Questionnaire-4 score difference-in-difference estimate, 0.11 points).

Meaning Living in states with trigger abortion bans compared with living in states without such bans was associated with a small but significantly greater increase in anxiety and depression symptoms after the *Dobbs* abortion opinion.

an internet-based Qualtrics data collection platform. Census bureau-generated person-level weights adjusted for nonresponse, adults per household, and coverage primarily via benchmarking to the American Community Survey (ACS).⁹ The mean population-weighted response rate to the HPS overall was 6.04% during the study period, consistent with the Census Bureau's goal of 5%.^{10,11}

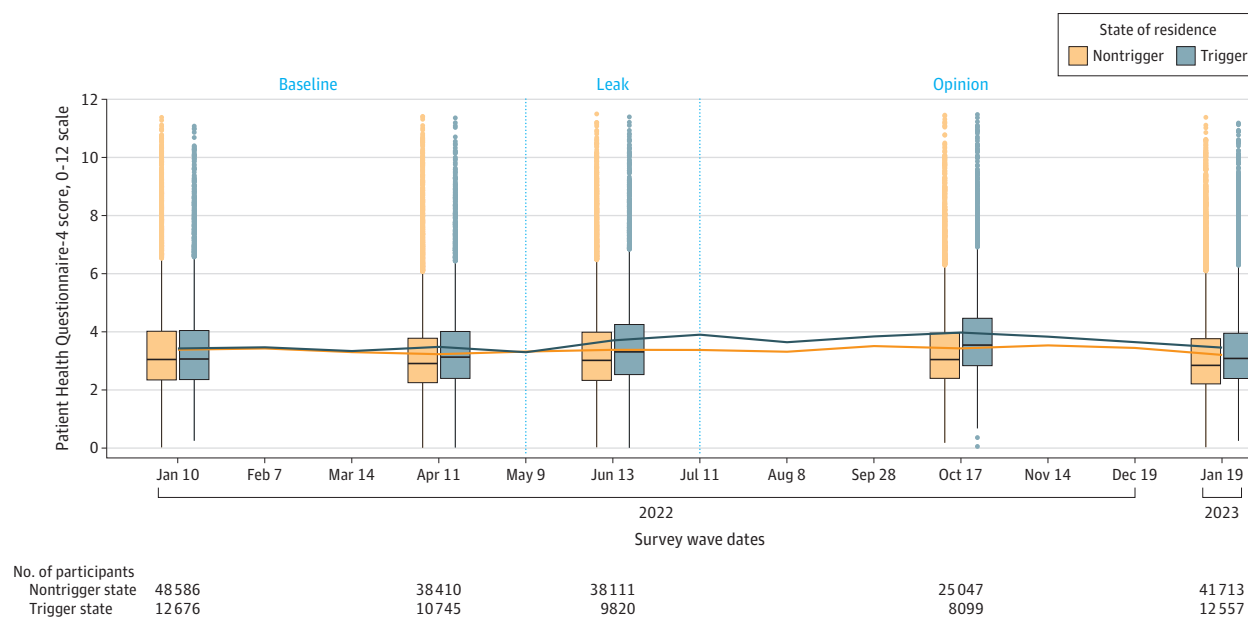
Exposure and Other Measures

To define the exposure and comparison groups, we created a variable indicating whether respondents resided in 1 of the 13 states with a trigger law (Arkansas, Idaho, Kentucky, Louisiana, Mississippi, Missouri, North Dakota, Oklahoma, South Dakota, Tennessee, Texas, Utah, and Wyoming), or in any of the other 37 nontrigger states across 3 distinct periods: (1) following the *Dobbs* decision on June 24, 2022 (the opinion), (2) between May 2, 2022, when the opinion was leaked (the leak) and when the *Dobbs* decision was released, and (3) before the leak (baseline). Respondents' state of residence was imputed from the Census Bureau's Master Address File. Key demographic variables included an indicator for respondent sex (female or male; gender was not collected in the survey), age (measured continuously), and self-reported race (Asian, Black, White, and other), and ethnicity (Hispanic vs non-Hispanic). The inclusion of race and ethnicity was informed because abortion access has historically differed by race and ethnicity.¹²

Outcome

The HPS contains the Patient Health Questionnaire-4 (PHQ-4), a well-established and validated screening tool for anxiety and depression.^{8,13} The screener asks respondents to report the frequency during which they felt down, worried, generally uninterested, and anxious over the past 2 weeks, as assessed by a 4-level scale ranging from "not at all" (score, 0) to "nearly every day" (score, 3). Responses across these 4 questions were aggregated to create a continuous 0 to 12 index (the standard measure of the PHQ-4). Scores ranging from 0 to 2 represent no symptoms of anxiety or depression; 3 to 5, mild; 6 to 8, moderate; and 9 to 12, severe.¹⁴ A PHQ-4 score of at least 6 typically indicates need for clinical intervention and is associated

Figure 1. Adjusted Patient Health Questionnaire-4 Score in the US Population



Regression models were used to estimate the population mean of the Patient Health Questionnaire-4 (PHQ-4) score within each of the 13 survey wave dates during the study period. PHQ-4 scores range from no to severe symptoms of anxiety or depression: none (0-2); mild (3-5), moderate (6-8), and severe (9-12). Vertical dotted lines mark the baseline period (ie, before the *Dobbs* decision), the leak period from May 2, 2022, until the decision, and the opinion period

after the decision was released on June 24, 2022. See Figure 2 for the list of trigger states. Box plots reflect the unweighted distribution of PHQ-4 scores in quartiles. Boxes indicate the first, median, and third quartiles; whiskers, the furthest value with $1.5 \times$ IQRs of the first and third quartile, and dots, outside values. The number of participants represent the unweighted sample size.

with increased health care use, disability days, and functional impairment.¹⁵ The minimal clinically important difference of the PHQ-4 scale is not known.

Statistical Analysis

To describe the sample, we computed the survey weighted means of all variables and their unweighted frequencies. A difference-in-differences approach was used to estimate the changes in mean PHQ-4 score among residents of trigger states between the leak and baseline, and the opinion and baseline periods, respectively, relative to the same changes from baseline among those living in nontrigger states. This modeling strategy enabled us to estimate the changes in symptoms of anxiety and depression between the baseline period and after the opinion, and separately, between the baseline period and after the widely publicized leak that preceded the opinion. Estimation was via population-weighted least squares, and standard errors were clustered by state.

We estimated separate difference-in-differences models for females aged 18 through 45 years and similarly aged males as a comparator, and conducted a test of interaction between males and females aged 18 through 45 years. To be thorough, we also estimated models for males and females older than 45 (eExhibit in Supplement 1). All reported results are unadjusted for covariates because the inclusion of covariates in difference-in-differences models can lead to inaccurate results¹⁶; however, estimates adjusted for the variables described in eTable 1 in Supplement 1 were similar and are reported in eTable 2 in Supplement 1.

Difference-in-differences models require the data to be consistent with the parallel-trends assumption, which assumes that trends in anxiety and depression symptoms would have evolved similarly in the trigger and nontrigger states had *Roe* not been overturned. Although this counterfactual is not directly testable, we examined differences in mean PHQ-4 scores between trigger and nontrigger states during the baseline period to provide supporting evidence of the lack of difference in baseline trends, which we would expect to have persisted in the postleak and postopinion periods in the absence of the *Dobbs* decision.¹⁵ These trends are depicted in Figure 1. A quantitative assessment of differences in the baseline period is reported in eTable 3 in Supplement 1. Similar trends are depicted for males and females aged 18 through 45 years in eFigures 1 and 2 in Supplement 1, and are further explored in eFigure 3 in Supplement 1. Difference-in-differences models can be biased by heterogeneities in treatment timing, so stacked difference-in-differences models, which incorporate the nontrigger states that passed abortion bans following the opinion, were estimated as a sensitivity test.¹⁷ To address the concern that the nontrigger states are not an appropriate comparison group, 2 additional sensitivity analyses were conducted. First, a synthetic control analysis, which systematically selects a weighted mean of comparison states to closely match the preperiod trends in a given trigger state, was conducted. Second, a model omitting Texas was estimated to account for the possibility that the population of Texas, relative to the other 12 trigger states, would drive results. To further explore the clinical relevance of the results, we estimated

Table. Characteristics of the 2022 Waves of the Household Pulse Survey

Characteristic	Percent of respondents ^a					
	Baseline (before May 2, 2022) ^b		Leak (May 2, 2022-June 23, 2022) ^b		Opinion (June 24, 2022, and after) ^b	
	Trigger state (n = 55 993) ^c	Nontrigger state (n = 208 375)	Trigger state (n = 22 882) ^c	Nontrigger state (n = 85 063)	Trigger state (n = 80 979) ^c	Nontrigger state (n = 265 461)
Sex						
Male	49	48	49	49	49	48
Female	51	52	51	51	51	52
Age, y						
18-30	18	16	18	15	19	17
31-45	28	27	29	28	28	27
46-60	25	26	24	26	25	26
61-75	23	25	24	25	23	26
76-88	5	6	5	6	4	5
Race						
Asian	4	7	3	6	3	6
Black	11	11	11	11	11	11
White	80	77	80	77	80	76
Any other or combination ^d	5	5	6	6	6	6
Hispanic ethnicity	19	16	18	16	19	15
PHQ-4 score mean (SD) ^e	3.51 (3.77)	3.31 (3.64)	3.60 (3.79)	3.31 (3.62)	3.81 (3.80)	3.49 (3.64)
>5 clinical threshold	25	23	27	23	28	25
Severity						
Typical (0-2)	50	52	49	52	46	49
Mild (3-5)	25	25	24	25	27	26
Moderate (6-8)	12	11	13	11	12	12
Severe (9-12)	14	12	14	12	15	13

Abbreviation: PHQ-4, Patient Health Questionnaire-4.

^a The frequencies are unweighted, but the proportions are population weighted.^b For definitions of the baseline, leak, and opinion periods, see the Figure 1 legend.^c For the list of states that passed trigger abortion bans, see Figure 2.^d Represents American Indian or Alaska Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander individuals or any combination and are consolidated by the Household Pulse Survey as "other" for public use.^e PHQ-4 is measured continuously on a 0 to 12 scale, has a clinical threshold of 6 or greater, and has clinically important cut points corresponding to increasing severity of symptoms at scores of 2, 5, and 8.

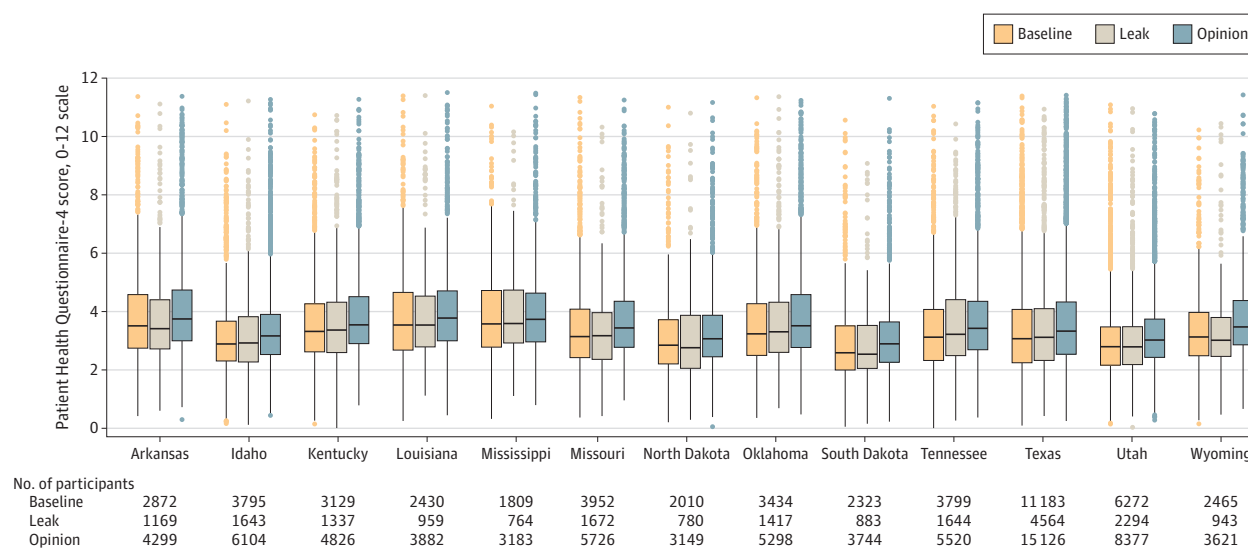
additional difference-in-differences models of exceeding key points along the range of PHQ-4, namely, mild (3-5), moderate (6-8), or severe (9-12) symptoms of anxiety and depression. All analyses were conducted in R version 4.1.0. All statistical testing was 2-sided at a significance threshold of .05.

Results

The mean population-weighted response rate to the HPS was 6.04% during the study period. The analytic sample for this study included all respondents who answered the items that comprise the PHQ-4 survey, which included about 87% of the HPS sample (N = 718 753). The population-weighted mean age was 48 years (SD, 17 years), and 51% were female. The Table contains key population-weighted descriptive statistics and unweighted counts of the sample among trigger and nontrigger state respondents across the baseline, leak, and opinion periods. Comparisons of population-weighted demographic statistics across our analytic sample, the HPS sample, and the 2022 ACS indicate that our inclusion criterion did not harm population representativeness (eTable 4 in Supplement 1). Overall,

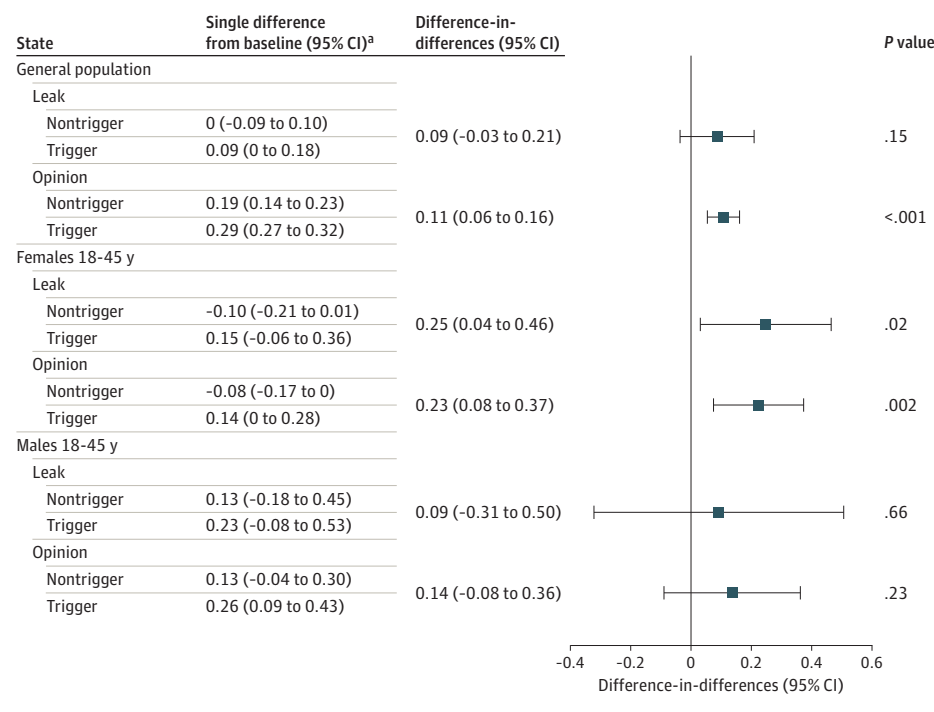
there were 159 854 respondents living in trigger states and 558 899 living in nontrigger states. The population-weighted mean PHQ-4 score in nontrigger states was 3.31 (95% CI, 3.27-3.34) at baseline, 3.31 (95% CI, 3.25-3.37) after the leak, and 3.49 (95% CI, 3.46-3.53) after the opinion; in trigger states, it was 3.51 (95% CI, 3.44-3.59), 3.60 (95% CI, 3.49-3.72), and 3.81 (95% CI, 3.75-3.87), respectively. The Table displays the population-weighted proportion of females within trigger and nontrigger states across all periods, as well as the distribution of age, race, and ethnicity. Characteristics such as the distribution of income, education, and the number of children in the household are displayed in eTable 2 in Supplement 1. Figure 2 displays boxplots that describe the distribution of PHQ-4 scores within each of the 13 trigger states.

Figure 3 displays results of the difference-in-differences analysis. Overall, residing in a trigger state compared with a nontrigger state was significantly associated with a greater increase in symptoms of anxiety and depression from baseline to after the *Dobbs* opinion, measured by a greater mean increase in the PHQ-4 score of 0.11 (95% CI, -0.06 to 0.16; $P < .001$) among residents of trigger states relative to the change in PHQ-4 observed during the same period among residents

Figure 2. Distribution of Patient Health Questionnaire-4 Score of the Trigger States Before and After the *Dobbs* Leak and Opinion

See the legend in Figure 1 for the definitions of baseline, leak, and opinion periods. Box plots reflect the unweighted distribution of Patient Health Questionnaire-4 (PHQ-4) scores in quartiles. See Figure 1 legend for PHQ-4

score ranges. The boxes indicate the first, median, and third quartiles; whiskers, extend to the furthest value with $1.5 \times$ IQRs of the first and third quartile; and dots, outside values. The sample size is unweighted.

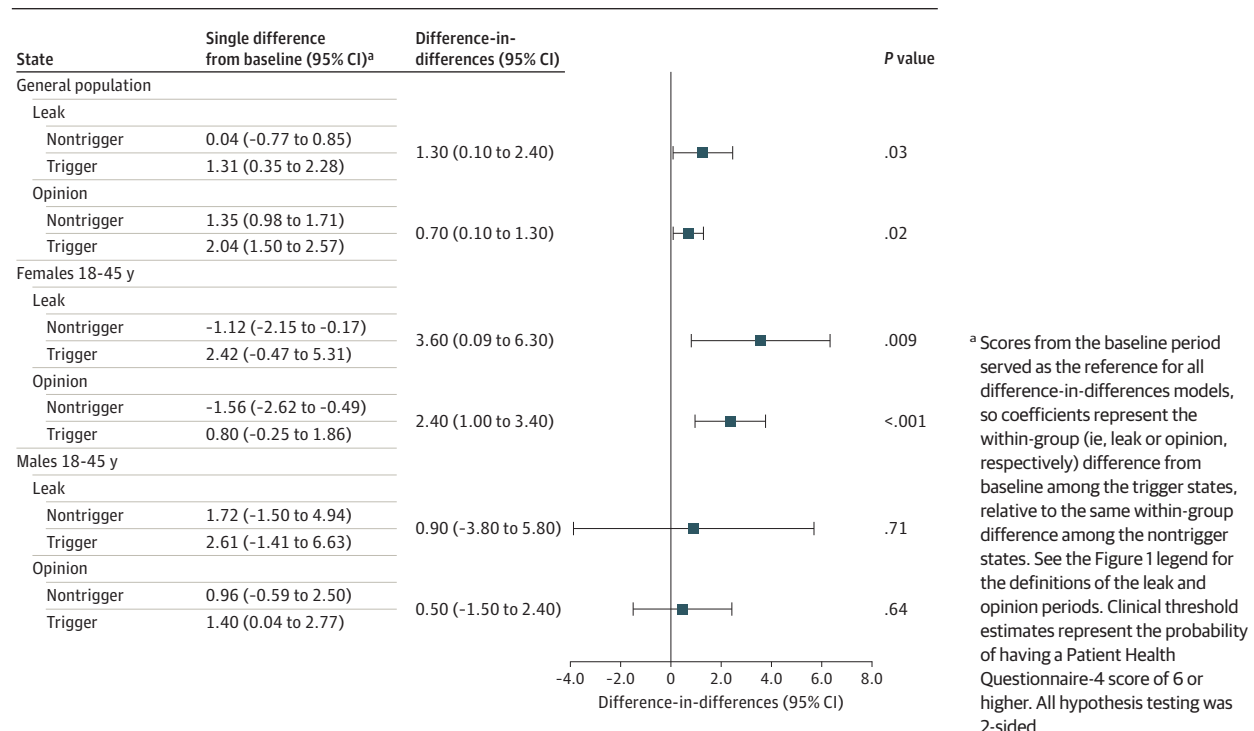
Figure 3. Population-Mean Patient Health Questionnaire-4 Before and After the *Dobbs* Leak and Opinion in Trigger vs Nontrigger States

^a Scores from the baseline period before the *Dobbs* decision leak served as the reference for all difference-in-differences models, so coefficients represent the within-group (ie, leak or opinion, respectively) difference from baseline among the trigger states, relative to the same within-group difference among the nontrigger states. All hypothesis testing was 2-sided. See the Figure 1 legend for the definitions of the leak and opinion periods.

of states without trigger laws. From baseline to after the draft was leaked, the change in PHQ-4 was not significantly different for those in trigger vs nontrigger states (difference-in-differences estimate, 0.09; 95% CI, -0.03 to 0.21; $P = .15$).

Among females aged 18 through 45 years, residing in a trigger state compared with a nontrigger state was significantly as-

sociated with a greater increase in the mean PHQ-4 score of 0.23 (95% CI, 0.08 to 0.37; $P = .002$) from baseline to after the opinion and was also significantly associated with a greater mean increase of 0.25 (95% CI, 0.04 to 0.46; $P = .02$) from baseline to after the leak. Among males aged 18 through 45 years, the difference-in-difference estimates were not significantly different

Figure 4. Probability of Exceeding the Patient Health Questionnaire-4 Clinical Threshold Before and After the *Dobbs* Leak and Opinion

for the analyses comparing the baseline period to after the opinion (difference-in-differences estimate, 0.14; 95% CI, -0.08 to 0.36; $P = .23$) or for the analyses comparing the baseline period to after the leak (difference-in-differences estimate, 0.09; 95% CI, -0.31 to 0.50; $P = .66$). The tests for interaction suggested that the difference-in-differences estimates were statistically significantly different for females vs males aged 18 through 45 years for the baseline to postopinion analysis (P for interaction = .02) but not for the baseline to the postleak analysis (P for interaction = .20). eTable 5 in [Supplement 1](#) contains population mean PHQ-4 scores and their 95% CIs across the baseline, leak, and opinion periods among the general population of those living in nontrigger states and trigger states and among the subgroup of females aged 18 through 45 years ($n = 153\,108$) and similarly aged males ($n = 102\,581$) that underpin the difference-in-differences analysis.

Figure 4 reports the estimates of difference-in-differences models focused on the probability of exceeding the PHQ-4's clinical threshold. From baseline to after the opinion, residing in a trigger vs a nontrigger state was significantly associated with a 0.7 percentage point (95% CI, 0.11 to 1.28; $P = .02$) greater increase in the probability of exceeding the PHQ-4's clinical threshold. From baseline to after the leak, residing in a trigger vs a nontrigger state was significantly associated with a 1.27 percentage point greater increase (95% CI, 0.12 to 2.43; $P = .03$). Among females aged 18 through 45 years, living in trigger vs nontrigger states was significantly associated with a 2.36-percentage point (95% CI, 0.96 to 3.76; $P < .001$) greater increase in the probability of exceeding the PHQ-4's clinical threshold from baseline to after the opinion, and with a 3.58 percentage point (95% CI, 0.81 to 6.35; $P = .009$) greater in-

crease in the probability of exceeding the PHQ-4's clinical threshold from baseline to after the leak. Among males aged 18 through 45 years living in trigger vs nontrigger states, there was no significant difference in the change in the probability of exceeding the PHQ-4's clinical threshold after the leak (0.90 percentage points; 95% CI, -3.89 to 5.69; $P = .71$). The same held true for the opinion period (0.46 percentage points; 95% CI, -1.50 to 2.41; $P = .64$). eTable 6 in [Supplement 1](#) displays the population proportion of individuals exceeding the PHQ-4's clinical threshold across the baseline, leak, and opinion periods among residents in nontrigger and trigger states in the general population and among the subgroups of males and females aged 18 through 45 years, respectively.

eFigure 4 in [Supplement 1](#) reports the results of difference-in-differences models of the relationship between the *Dobbs* decision and the distribution of PHQ-4 scores, defined by having mild, moderate, or severe symptoms of anxiety or depression. For the analyses comparing the baseline period to after the *Dobbs* opinion, living in a trigger vs a nontrigger state was significantly associated with a greater increase in the probability of having both severe (difference-in-differences estimate, 0.01 percentage point; 95% CI, 0.00 to 0.01; $P < .001$) and mild (difference-in-differences estimate, 0.01 percentage point; 95% CI, 0.00 to 0.02; $P = .04$) symptoms of anxiety or depression but was not associated with any significant difference in the change in the probability of having moderate symptoms (difference-in-differences estimate, -0.002 percentage points; 95% CI, -0.01 to 0.00; $P = .43$).

eTable 7 in [Supplement 1](#) presents estimates from stacked difference-in-differences models, which defined the exposed group as the states that implemented an abortion ban

at any time during the study period. Estimates of these models and the core difference-in-differences models were similar. There was a greater worsening of the mean PHQ-4 score for those in states with abortion bans than among those in states with no abortion bans by 0.11 (95% CI, 0.05 to 0.17; $P < .001$) in the general population and 0.17 (95% CI, 0.08 to 0.26; $P < .001$) among females aged 18 through 45 years. Estimates for males aged 18 through 45 years showed no significant difference in the change in the mean PHQ-4 scores for trigger vs nontrigger states (difference-in-differences estimate, 0.11; 95% CI, -0.07 to 0.29). eFigure 5 in Supplement 1 presents the results of a synthetic control analysis in which the counterfactual that Texas had never passed a trigger law was imputed using a weighted average of characteristics from other states. The difference in mean PHQ-4 scores between the observed Texas and the synthetic Texas were similar at baseline and rose significantly following the leak and the opinion, potentially indicating worsened anxiety and depression symptoms following the ruling ($P = .03$). eTable 8 in Supplement 1 contains the results of difference-in-differences models omitting Texas from the analysis, which considers the possibility that Texas, as the state with the largest population in the set of exposed states, explains most of the variation observed in the core models. The 95% CIs of the models omitting Texas and the core models overlap, which supports the robustness of the core results.

Discussion

Using a nationally representative dataset and variation in trigger law status across states, this study demonstrated that living in states with trigger abortion bans, compared with living in states without such bans, was associated with a small but significantly greater increase in PHQ-4 scores after the *Dobbs* opinion. The study also demonstrated an increased probability of surpassing the PHQ-4's threshold for clinically relevant symptoms in states with trigger abortion bans relative to states without such bans after the *Dobbs* opinion and its leak.

Visual inspection of the plots of mean PHQ-4 score suggest that depression and anxiety symptoms worsened, overall, after the leak of the *Dobbs* opinion with continued worsening after the decision was released. Although this overall worsening peaked in October 2022 and then attenuated, the gap between trigger and nontrigger states, the main target of analysis, persisted. Females aged 18 through 45 years, an age cutoff selected for consistency with measures of the lifetime incidence of abortion, faced greater worsening of anxiety and depression symptoms in trigger vs nontrigger states, whereas males of a similar age experienced minimal or negligible changes, a pattern also detected in analyses of the PHQ-4's clinical threshold. Additionally, distributional analyses revealed a significantly increased change in the probability of having severe symptoms of anxiety or depression after the *Dobbs* opinion for those in trigger states compared with those in nontrigger states. These findings were robust to various specifications of the exposure, varying of model assumptions, and alternative statistical designs.

Although the results were statistically significant, they were small in magnitude. This is unsurprising for a study of the association of a distal factor—a policy change—with self-reported mood symptoms at the population level.

The findings provide new evidence about the relationship between the changing abortion policy landscape and mental health following the *Dobbs* opinion. Although there were increases in symptoms of anxiety and depression for the general population after the opinion, changes in symptoms of anxiety and depression were greater among those living in states with trigger abortion bans, and in particular, among females within the age range generally used to compute lifetime abortion incidence. This finding could be related to many factors, including fear about the imminent risk of abortion denial; uncertainty around future limitations on abortion and other related rights, such as contraception; worry over the ability to receive lifesaving medical care during pregnancy; and a general sense of violation and powerlessness related to loss of the right to reproductive autonomy.

Limitations

This study has several limitations. First, the HPS is a relatively new data set developed for COVID-19 research and has been used less frequently in research than other data produced by the Census Bureau. However, these data benefit from the Census Bureau's weighting scheme and sampling frame.¹⁵ Second, the data were pooled cross sections of different individuals over time, not a panel of the same sample repeatedly measured, which makes it more difficult to adjust for individual characteristics. The large sample size, population representativeness, and state-level analyses help to ameliorate this concern. Third, the population-weighted response rate of the HPS was around 6% over the study period. Although some degree of nonresponse bias and variance is a feature of almost all statistical surveys, the Census Bureau used several techniques to mitigate bias, including constructing sample weights that adjusted for key differences in geography and demographic characteristics. After weighting, the Census Bureau's analysis found similar observable characteristics between the HPS sample and the ACS. Additionally, the Census Bureau designed the study around an expected 5% response rate.¹² Fourth, the HPS did not capture individual-level political ideology, which likely plays a role in how the *Dobbs* decision operates on mental health, although the modeling approach in this study does account for state-level variation in political ideology. As a result, this study was unable to evaluate whether the *Dobbs* decision was associated with positive changes in mental health among individuals with political ideology aligned with more restrictions on abortion access.

Conclusion

In this study of US survey data from December 2021 to January 2023, residence in states with abortion trigger laws, compared with residence in states without such laws, was associated with a small but significantly greater increase in symptoms of anxiety and depression after the *Dobbs* opinion.

ARTICLE INFORMATION

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Acquisition, analysis, or interpretation of data: Thornburg, Rosen, Eisenberg.
Drafting of the manuscript: Thornburg, Rosen.
Critical review of the manuscript for important intellectual content: All authors.
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